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Metal encapsulated ceramic tile thermal insulation, and associated systems and methods

Abstract

A metal encapsulated ceramic tile thermal insulation system for rockets and associated methods is disclosed. A representative system includes a launch vehicle having a first end and a second end generally opposite the first end and includes a heat shield positioned at the second end. The heat shield includes a plurality of thermal protection apparatuses, where individual of the thermal protection apparatuses include ceramic tiles encapsulated by inner and outer metal layers, which are positioned on opposing top and bottom surfaces of the ceramic tiles. The plurality of thermal protection apparatuses includes a plurality of pins positioned within corresponding holes drilled through the ceramic tiles and are secured to the metal layers. The outer metal layer can protect the ceramic tile from tool strikes and debris and can also prevent water from reaching and being absorbed by the ceramic tile.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a divisional of U.S. patent application Ser. No. 16/437,529, titled "METAL ENCAPSULATED CERAMIC TILE THERMAL INSULATION, AND ASSOCIATED SYSTEMS AND METHODS," and filed Jun. 11, 2019, which claims the benefit of and priority to U.S. Provisional Patent Application No. 62/684,145, titled "METAL ENCAPSULATED CERAMIC TILE THERMAL INSULATION,

TECHNICAL FIELD

(1) The present technology relates to metal encapsulated ceramic tile thermal insulation, and associated systems and methods, for example, thermal protection systems and heat shields for rockets.

BACKGROUND

(2) Rocket manufacturers continually strive to reduce the costs of launching a payload into space. One approach for reducing such costs is to retrieve one or more booster stages used to propel the space launch vehicle. In a particular approach, the booster is launched and landed vertically and refurbished for another launch. One drawback to this approach is that the exterior surfaces of the booster, including the engine nozzles, are subjected to high temperatures, which can result in damage to these surfaces during ascent and/or descent. To overcome this drawback, launch and reentry vehicle manufacturers utilize insulation and cooling systems designed to reduce the amount of heat the engine nozzles and/or other surfaces are exposed to during flight. Conventional types of insulation include ceramic tiles that form a heat shield on the bottom surface of the booster. However, these ceramic tiles are brittle and not very robust, often requiring refurbishment between launches. Further, the ceramic tiles are typically very porous and must be waterproofed before every launch to prevent the tiles from soaking up water, which undesirably increases the weight of the booster. Accordingly, there is a need for improved insulation systems, e.g., for reusable launch vehicles.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a partially schematic, side elevation view of a representative launch vehicle system having insulation systems configured in accordance with embodiments of the present technology.
- (2) FIG. 2 is an enlarged, partially schematic, elevation view of one end of the launch vehicle of FIG. 1 having a heat shield formed from thermal protection apparatuses, configured in accordance with embodiments of the present technology.
- (3) FIG. 3 is a perspective view of one of the thermal protection apparatuses shown in FIG. 2.
- (4) FIG. 4 is a cut-away isometric view of a representative thermal protection apparatus, configured in accordance with embodiments of the present technology.
- (5) FIGS. 5 and 6 show schematic, cross-sectional views of representative thermal protection apparatuses, configured in accordance with embodiments of the present technology.
- (6) FIGS. 7 and 8 show schematic, cross-sectional views of sealing mechanisms between two adjacent thermal protection apparatuses, configured in accordance with embodiments of the present technology.

DETAILED DESCRIPTION

(7) Several embodiments of the present technology are directed to systems and apparatuses for insulating structures, e.g., rocket structures and components, to reduce the effects of heat. For example, the present technology can include a thermal protection apparatus having a rigid insulation layer positioned between and attached to two metal layers using pins. The thermal protection apparatus can be attached to the rocket components with an adhesive or with fasteners that couple to the pins. This approach can combine the insulating properties of the rigid insulation layer with the strength and durability of the metal to insulate the rocket when the rocket reenters the atmosphere and lands. For example, the rigid insulation can inhibit or prevent heat from penetrating through the thermal protection apparatus to the body of the rocket during reentry, while the metal layers can reduce damage to the rigid insulation from foreign object damage and can help

to prevent water from reaching and being absorbed by the insulation. Further, the pins can be placed in holes formed through both the insulation and the metal layers and can be attached to the metal layers in order to secure the metal layers to the insulation. As a result, the thermal protection apparatus can insulate the rocket structures and components using the insulation while the metal layers can protect the insulation from damage.

(8) Specific details of several embodiments of the disclosed technology are described below with reference to particular, representative configurations. The disclosed technology can be practiced in accordance with rockets, heat shields, and/or insulation having other suitable configurations.

Specific details describing structures or processes that are well-known and often associated with rockets and insulation, but that can unnecessarily obscure some significant aspects of the presently disclosed technology, are not set forth in the following description for purposes of clarity.

Moreover, although the following disclosure sets forth some embodiments of different aspects of the disclosed technology, some embodiments of the technology can have configurations and/or components different than those described in this section. Further, unless otherwise specifically noted, elements depicted in the drawings are not necessarily drawn to scale. As such, the present technology can include some embodiments with additional elements and/or without several of the elements described below with reference to FIGS. 1-8.

(9) FIG. 1 is a partially schematic elevation view of a system **100** that includes a launch vehicle **110** having opposing first and second ends **111** and **112** and that is capable of ascending and descending in opposite directions but with generally the same orientation (e.g., a tail-down orientation).

Accordingly, the launch vehicle **110** can ascend in a first direction **101** and descend in a second direction **102**. The launch vehicle **110** includes a first stage **113** (e.g., a booster) having a body **115**, and a second stage **114** that can carry a payload **116** (e.g., one or more humans, supplies, experiments, etc.). In other embodiments, the launch vehicle **110** can include other numbers of stages (e.g., a single stage or more than two stages). The launch vehicle **110** is elongated along a vehicle axis **V** and the payload **116** can be positioned at the first end **111**. The launch vehicle **110** can include a pre-determined separation location **117** at which the first stage **113** separates from the second stage **114**, typically during ascent. The launch vehicle **110** can also include landing gear that support the launch vehicle **110** when it is not airborne or space-borne, and one or more elements (e.g., fins) that provide stability and control for the launch vehicle **110**.

(10) The first stage **113** includes a propulsion system **118** positioned at the second end **112** and coupled to the body **115**. The propulsion system **118** includes nozzles **119** oriented to direct exhaust products in a generally downward direction (i.e. in the second direction **102**). The propulsion system **118** also includes a plurality of combustion chambers located within the body **115** of the launch vehicle **110**, with each of the nozzles **119** coupled to a given one of the combustion chambers. Each of the combustion chambers receives fuel from a fuel pump coupled to a fuel tank within the body **115**. An igniter ignites the fuel within the combustion chambers, creating high energy exhaust products that are directed through the associated nozzles **119**. Each of the nozzles **119** is positioned to direct the exhaust products away from the second end **112** of the launch vehicle **110** (e.g., in the second direction **102**), thereby generating thrust that propels the launch vehicle in the first direction **101**.

(11) Once the launch vehicle **110** reaches a specific and pre-determined point in the launch process (e.g., a specific altitude or speed, a specific amount of fuel consumed, etc.), the first and second stages **113** and **114** separate from each other at the separation location **117**. In some embodiments, the second stage **114** includes a secondary propulsion system used to propel the second stage **114** towards its final destination after the first and second stages **113** and **114** separate, while the first stage **113** returns back to earth. In other embodiments, the second stage **114** does not include a secondary propulsion system and both the first and second stages **113** and **114** return to earth after separation. The first stage **113** can also include lateral thrusters to stabilize and control the first stage **113** as it returns to earth. Further details of the lateral thrusters are included in pending U.S.

(12) As the first stage **113** descends, the propulsion system **118** and the lateral thrusters work together to control the orientation and speed of the first stage **113** as it returns to earth. In a representative embodiment, the propulsion system **118** and the lateral thrusters control the first stage **113** as it descends such that first stage **113** moves in the second direction **102** and the vehicle axis **V** is generally parallel to the second direction **102**. As it approaches the landing site, the first stage **113** has a generally vertical orientation such that the second direction **102** and the vehicle axis **V** are both oriented perpendicular to the ground and the one or more nozzles **119** direct the exhaust products downward, causing the first stage **113** to decelerate. Landing gear, which can be stowed during ascent and descent, extend from the body of the first stage **113** and support the weight of the first stage **113** as it lands. Once the first stage **113** lands, the propulsion system **118** shuts down and the first stage **113** is secured to the landing site. In this way, the first stage **113** may be used for subsequent launches and only minor refurbishments and part replacements may be required between subsequent launches of the first stage **113**.

(13) Throughout the launching and landing processes, the launch vehicle **110** is subjected to extreme conditions. For example, the second end **112** of the first stage **113** is subjected to high air pressures and temperatures caused by friction between the air and the second end **112**. To reduce the effects of the high temperatures, the launch vehicle **110** includes a thermal protection system **120** positioned at or at least toward the second end **112**, and that includes shielding, insulation, and/or other cooling systems. For example, to protect the second end **112** of the body **115** from these high temperatures, the thermal protection system **120** also includes a heat shield **121** coupled to the body **115** at the second end **112**. The heat shield **121** is positioned against the body **115** and insulates the body **115** from the high temperatures.

(14) FIG. 2 shows an enlarged view of the second end **112** of the first stage **113** of the launch vehicle **110**. The heat shield **121** is coupled to the body **115** of the first stage **113** adjacent to the nozzles **119** and is used to insulate the body **115** from high temperatures as the first stage **113** descends in the second direction **102**. The heat shield **121** can be formed from a plurality of thermal protection apparatuses **130** securely coupled to the body **115**. The thermal protection apparatuses **130** can be modular, and can be arranged in an array over the second end **112** of the body **115** to fully cover the second end **112**. In some embodiments, each of the thermal protection apparatuses **130** can be generally rectangular and planar. In other embodiments, some of the thermal protection apparatuses **130** can be generally rectangular and planar while others can be curved and/or rounded to conform to the shape of the second end **112**.

(15) FIG. 3 shows a perspective view of a representative rectangular thermal protection apparatuses **130** having a length **L** and width **W**. In some embodiments, the thermal protection apparatus **130** can be sized and shaped such that the length **L** is approximately 18 inches and the width **W** is approximately 12 inches. In other embodiments, the length **L** and width **W** can have other suitable values. The thermal protection apparatus **130** can include a ceramic tile **134** positioned between an outer metal portion (e.g., a layer) **133** and an inner metal portion (e.g., a layer) **135**. With this arrangement, the outer metal layer **133** defines an outer surface **131** of the thermal protection apparatus **130** while the inner metal layer **135** defines an inner surface **132**. When the thermal protection apparatus **130** is coupled to the launch vehicle, the thermal protection apparatus **130** is positioned such that the inner surface **132** faces towards the body of the launch vehicle while the outer surface **131** faces away. The thermal protection apparatus **130** can also include one or more holes **144** that extend through the outer metal layer **133**, the ceramic tile **134**, and the inner metal layer **135**. In representative embodiments, the metal layers **133** and **135** and the ceramic tile **134** are coupled together with one or more corresponding pins **136** positioned within the holes **144** and coupled to the metal layers **133** and **135**. The pins **136** can be arranged in rows and columns within the thermal protection apparatus **130** and each of the pins can be configured to extend from the outer surface **131** to the inner surface **132** by passing through the outer metal layer **133**, the ceramic

tile **134**, and the inner metal layer **135**. The pins **136** are each securely coupled to the outer metal layer **133** to hold the outer metal layer **133** against the ceramic tile **134** and to prevent the outer metal layer **133** from detaching. The thermal protection apparatus **130** can also include edging **137** positioned around the perimeter of the ceramic tile **134**.

(16) FIG. 4 shows an isometric cut-away view of a representative thermal protection apparatus **130**. The ceramic tile **134** includes opposing top and bottom surfaces **140** and **141**. The outer and inner metal portions (e.g., layers) **133** and **135** can each include a thin layer of metal and can be positioned directly against the ceramic tile **134** such that the outer metal layer **133** is positioned against the top surface **140** and the inner metal layer **135** is positioned against the opposing bottom surface **141**. The ceramic tile **134** is generally rectangular and can include a notch **138** formed in the top surface **140** and extending around the perimeter of the ceramic tile **134**. The edging **137** can be formed from a thin sheet of corrugated metal (e.g., Inconel® 660 available from Special Metals Corporation at www.specialmetals.com/) positioned directly against edge portions **142** of the ceramic tile **134** and coupled to the outer and inner metal layers **133** and **135**. As will be discussed in greater detail below, the outer metal layer **133** can laterally expand when the thermal protection apparatus **130** is heated, while the ceramic tile **134** and the inner metal layer **135** may not expand (or may not expand significantly). The corrugated structure of the edging **137** can increase the flexibility of the edging **137** to allow the edging **137** to remain coupled to both the outer metal layer **133** and the inner metal layer **135** as the outer metal layer **133** expands.

(17) The edging **137** can include top and bottom bent portions **143a** and **143b**, which are bent over the edge portions **142** of the ceramic tile **134**. More specifically, the top bent portion **143a** is positioned between the outer metal layer **133** and the ceramic tile **134** in the notch **138** while the bottom bent portion **143b** is positioned against the inner surface **132** such that the inner metal layer **135** is positioned between the bottom bent portion **143b** and the ceramic tile **140**. Further, the top bent portion **143a** is positioned within the notch portion **138** such that top bent portion **143a** is coplanar with the top surface **140**. As a result, the edging **137** does not extend above the top surface **140**, thereby allowing the outer surface **131** to remain generally flat. To couple the edging **137** to the outer and inner metal layers **133** and **135**, the outer metal layer **133** can be welded (e.g., tack welded) to the top bent portion **143a** and the inner metal layer **135** can be welded to the bottom bent portion **143b**. In some embodiments, the ceramic tile **134** can also include a second notch portion formed in the bottom surface **141** that extends around the perimeter of the ceramic tile **134** and the bottom bent portion **143b** can be positioned within the second notch portion. In these embodiments, the bottom bent portion **143b** can be positioned between the inner metal layer **135** and the ceramic tile **134** such that the edging **137** does not extend below the inner surface **132**. As will be discussed in further detail below, the outer metal layer **133** can include a lip portion **139** that extends past the edge portions **142** and the edging **137** and that can be used to form a seal between adjacent thermal protection apparatuses **130**.

(18) In representative embodiments, the outer metal layer **133** is formed from sheet metal that is less than or equal to 0.25 inches thick and that is cut into a desired size and shape. The outer metal layer can be formed from a metal having high strength and oxidation resistance at high temperatures. During reentry, the temperature that the outer metal layer **133** is heated to generally depends on the speed of the first stage as it descends, where the speed is generally dependent on the altitude at which the first and second stages separated. As such, the outer metal layer **133** can be formed from a metal that retains its strength and oxidation resistance throughout the descent. For example, in embodiments for which the expected temperature does not exceed 1400° F., the outer metal layer **133** can be formed from a metal such as titanium (or alloys that include titanium), which retains its strength and oxidation resistance up to approximately 1400° F. However, in embodiments for which the expected temperature reaches temperatures greater than 1400° F., the outer metal layer **133** can be formed from nickel-based alloys (e.g., Haynes 230® alloys available from Haynes International at www.haynesintl.com/, Inconel® 625 alloys available from Special

Metals Corporation at www.specialmetals.com/, HASTELLOY C-22® alloys available from Haynes International at www.haynesintl.com/, etc.), which can retain their strength and oxidation resistance at temperatures up to approximately 2000° F., or refractory alloys (e.g., TZM alloys available from Ed Fagan Inc. at www.edfagan.com/, C-103 alloys available from ATI at www.atimetals.com/), which can retain their strength at temperatures greater than 3000° F.

(19) The outer metal layer **133** typically does not reflect and/or reject heat incident on the outer surface **131**. Accordingly, when the outer metal layer **133** is heated, the heat passes through the outer metal layer **133** to the ceramic tile **134**. The ceramic tile **134** can be formed from a rigid and porous ceramic material having a low thermal conductivity, a high temperature resistance, and a low coefficient of thermal expansion, and that can include silica and/or alumina fibers bonded together (e.g., CT300 Tooling Board available from COMPOTOOL at www.compotool.com/). As such, when the ceramic material is exposed to heat and high temperatures, the ceramic tile **134** generally retains its size and shape while efficiently rejecting heat transfer. With this arrangement, points within the ceramic tile **134** near the top surface **140** can be hotter than points within the ceramic tile **134** further from the top surface **140**. As a result, a temperature gradient can be established through the ceramic tile **134**. Accordingly, the ceramic tile **134** can prevent heat from penetrating completely through the ceramic tile **134** so that temperatures at the bottom surface **141** are maintained at lower levels. Further, the ceramic material can have a low density and can be easily manufactured to have a suitable shape and a selected thickness T. The total amount of heat that the ceramic tile **134** rejects is at least partially dependent on the thickness T of the ceramic tile **134** and the thickness T can therefore be selected based on the amount of heat that the thermal protection apparatus **130** is expected to be exposed to during descent. In some embodiments, the thickness T can be approximately 1 inch, between 0.25 and 1 inch, or can be between 1 and 3 inches.

(20) In some embodiments, the porous ceramic material can be capable of readily absorbing water (e.g., water from the atmosphere such as rain, snow, humidity, etc. or water used in cooling or noise suppression systems). However, water is dense and saturating the ceramic tile **134** with water can increase the weight of the thermal protection apparatus **130**. To restrict or prevent water ingress, the outer metal layer **133** can act as a hermetic waterproof barrier and can prevent most of the water incident on the outer surface **131** from reaching and being absorbed by the ceramic tiles **134**. To further reduce the amount of water capable of being absorbed by the ceramic material, the thermal protection apparatus **130** can include waterproofing applied to the ceramic tile **134**. In some embodiments, the waterproofing can be applied to the ceramic tile **134** by submerging the ceramic tile **134** in the waterproofing material for a suitable amount of time.

(21) In some embodiments, the waterproofing can be applied to the bottom surface **141** and the edge portions **142** of the ceramic tile **134**. In other embodiments, the waterproofing can be applied through the entire thickness T. However, the high temperatures at points within the ceramic tile **134** near the top surface **140** can cause the waterproofing near the top surface **140** to burn off while the waterproofing near the bottom surface **141** can remain intact throughout the launch and landing. Because some of the waterproofing remains within the ceramic tile **134** and because the outer metal layer **133** can prevent most of the water from reaching the ceramic tile **134**, the amount of water that can be absorbed by the ceramic tile **134** can be reduced and the thermal protection apparatus **130** can be used for multiple launches and landings without having to apply waterproofing between launches.

(22) In addition to or in lieu of increasing the waterproofing abilities of the thermal protection apparatus **130**, the outer metal layer **133** can increase the durability and toughness of the thermal protection apparatus **130**. For example, the ceramic material can be brittle and can crack and break if struck by a tool or by foreign object debris (FOD) during flight. Accordingly, the outer metal layer **133** can provide protection to the ceramic tile **134** from tool strikes and FOD, thereby increasing the impact resistance of the thermal protection apparatus **130**. In this way, the ceramic

tile **134** can be formed from a wider array of materials. For example, the ceramic tile **134** can be formed from ceramic materials having a high temperature resistance but that are very brittle and tend to fracture easily, as the outer metal layer **133** can prevent the ceramic tile **134** from being struck by tool strikes and FOD. Furthermore, the outer metal layer **133** can be electrically conductive and can be capable of discharging electricity and/or avoiding charge build-up due to lightning, static charges, and/or other sources.

(23) Because the ceramic material can prevent or at least restrict heat from reaching the inner metal layer **135**, the inner metal layer **135** can be formed from different metals than the outer metal layer **133**. For example, in representative embodiments, the inner metal layer **135** is formed from aluminum or another suitable lightweight metal. The inner metal layer **135** can have a thickness of less than or equal to 0.25 inches and can be used as a back plate to help the pins **136** secure the outer metal layer **133** to the ceramic tile **134**. In other embodiments, however, the thermal protection apparatus **130** can be formed without the inner metal layer **135** and instead can include a layer of another suitable type of lightweight material, such as a composite, positioned against the bottom surface **141**. In still other embodiments, the thermal protection apparatus **130** can be formed without the inner metal layer **135** or any other layer such that the bottom surface **141** is exposed and the ceramic tile **134** is positioned directly against the body of the first stage. In these embodiments, the weight of the thermal protection apparatus **130** can be reduced compared to existing thermal protection systems, thereby reducing the cost of launching the launch vehicle.

(24) When heated, metal typically expands and can even deform. As such, when the thermal protection apparatus **130** is exposed to high temperatures, the outer metal layer **133** heats up and tends to expand laterally. However, the ceramic material that forms the ceramic tile **134** has a low coefficient of thermal expansion and the inner metal layer **135** is shielded from the high temperatures. Accordingly, neither the ceramic tile **134** nor the inner metal layer **135** expand significantly when the thermal protection apparatus **130** heats up, resulting in the outer metal layer **133** moving laterally relative to the ceramic tile **134** and/or the inner metal layer **135**. This thermal expansion mismatch can create stresses on the pins **136**, which extend through the entire thickness of the thermal protection apparatus **130**. To prevent the pins **136** from detaching from the outer metal layer **133** when the outer metal layer expands and moves, the pins **136** can be formed from a generally flexible material capable of bending and elastically deforming. For example, in some embodiments, the pins **136** can be formed from metals such as molybdenum or niobium. In other embodiments, the pins **136** can be formed from the same metal that the outer metal layer **133** is formed from. In this way, the pins **136** can remain securely attached to the outer metal layer **133** as the outer metal layer **136** expands and contracts.

(25) FIG. 5 shows a cross-sectional view of one of the pins **136** positioned within a hole **144**. To allow the pins **136** sufficient room to bend and flex, each of the holes **144** can be larger than the pins **136**. For example, in some embodiments, the pins **136** and the holes **144** can be generally cylindrical and the holes **144** can have a diameter larger than that of the pins **136**. In these embodiments, the pins **136** can bend and flex in any direction within the hole. In other embodiments, the pins **136** can be cylindrical while the holes **144** are slot-shaped such that the pins **136** are able to bend and flex along a predetermined direction. In general, the holes **144** can have any suitable size and shape that enables the pins **136** to bend and/or flex sufficiently during operation of the thermal protection apparatus **130**.

(26) Each of the holes **144** is formed through the outer metal layer **133**, the ceramic plate **134**, and the inner metal layer **135**, thereby extending from the outer surface **131** to the inner surface **132**. When forming the thermal protection apparatus **130**, the sheets of metal that form the outer and inner metal layers **133** and **135** can be positioned against the respective top and bottom surfaces **140** and **141** of the ceramic tile **134** before the holes **144** are formed. Once the metals layers **133** and **135** are properly positioned, a drill can be used to form the holes **144** through the outer metal layer **133**, the ceramic tile **134**, and the inner metal layer **135** and one of the pins **136** can be

positioned within each of the holes **144**. The pins **136** can be longer than the thickness of the thermal protection apparatus **130** such that the opposing ends of the pins extend beyond the outer and inner surfaces **131** and **132**. Once the pins **136** are positioned within their respective holes **144**, the pins **136** can be coupled to the outer and inner metal layers **133** and **135**. For example, the pins **136** can be securely held in place with threaded nuts, clips, fasteners, brackets, welds, and/or any combination thereof. In the illustrated embodiment, the pin **136** is spot-welded to the outer and inner metal layers **133** and **135** such that the pin **136** is attached to the outer metal layer **133** with weld **145a** and to the inner metal layer **135** with weld **145b**.

(27) After attaching the pins **136** to the inner and outer metal layers **133** and **135**, the thermal protection apparatus **130** can be attached to the body **115** of the first stage of the launch vehicle. In the illustrated embodiment, an adhesive **146** is applied to the inner surface **132** to attach the thermal protection apparatus **130** to the body **115**. The adhesive **146** can be a room temperature vulcanizing (RTV) silicone adhesive that can operate at temperatures up to 500° F. (e.g., RTV560 available from Momentive Performance Materials at www.momentive.com) or can include another suitable type of adhesive.

(28) In other embodiments, however, the thermal protection apparatus **130** can be attached to the body **115** without using an adhesive. For example, FIG. 6 shows a cross-sectional view of an embodiment of the thermal protection apparatus **130** that utilizes a fastener **147** coupled to the pin **136** to attach the thermal protection apparatus **130** to the body **115**. In the illustrated embodiment, the hole **144** extends through a portion (e.g., a layer) of the body **115**. The pin **136** is welded to the outer metal layer **133** with the weld **145a** and extends through the outer metal layer **133**, the ceramic tile **134**, the inner metal layer **135**, and a portion of the body **115**. The pin **136** extends completely through the portion of the body **115** and the fastener **147** attaches to the end of the pin **136** within the body **115**. In this way, the fastener **147** and the body **115** act as a bracket that uses the pin **136** and the weld **145a** to pull the outer metal layer **133** into the ceramic tile **134**. In other embodiments, the pin **136** can be attached to the body **115** using a different attachment mechanism, such as a nut. In still other embodiments, the pin **136** can be welded to the body **115**. In any of these embodiments, the portion of the body **115** to which the pin **136** is attached can be a shell that is accessible from inside for attachment. In other embodiments, the pin **136** can be attached using blind fastening techniques. In the illustrated embodiment, the inner metal layer **135** is positioned directly against the body **115** of the launch vehicle. In other embodiments, an adhesive (e.g., adhesive **146**) can be used such that thermal protection apparatus **130** is coupled to the body **115** with both the fastener **147** and the adhesive.

(29) When arranging the thermal protection apparatuses **130** on the launch vehicle, gaps between adjacent thermal protection apparatuses **130** may initially be present. For example, in some embodiments, adjacent thermal protection apparatuses **130** can be separated from each other by a gap of approximately 0.19 inches. To prevent water and heat from passing through these gaps, the gaps can be sealed. FIG. 7 shows a cross-sectional view of two thermal protection apparatuses **130** separated from each other by a gap **148**. In the illustrated embodiment, the upper metal layer **133** on the right thermal protection apparatuses **130** includes a lip portion **139**. The lip portion **139** extends over the gap **148** and is positioned over the outer surface **131** of the left thermal protection apparatus **130**. The lip portion **139** can be slightly bent so that it is positioned directly against the outer surface **131** and can be welded to the outer metal layer **133** of the left thermal protection apparatus **130** to couple the two thermal protection apparatuses **130** to each other. A gasket **149** can be positioned within the gap **148** between the two edgings **137**, and a ceramic rope seal **150** can be positioned between the gasket **149** and the body **115** of the launch vehicle.

(30) When the thermal protection apparatus **130** heats up, both the outer metal layer **133** and the edging **137** can expand due to thermal expansion. Accordingly, both of the outer metal layers **133** and both of the edgings **137** can expand into the gap **148**. As they expand, the width of the gap **148** decreases and the gasket **149**, which can be formed from a ceramic having a low thermal expansion

coefficient, can be pushed by one of the edgings **137** until it contacts the other edging **137**. The gasket **149** can be sandwiched between the two edgings **137**, forming a seal that prevents, or at least inhibits, heat from passing through the gap **148**. The ceramic rope seal **150** can be used to prevent, or at least inhibit, heat that passes by the gasket **149** from reaching the body **115**. Further, the temperature gradient established by the ceramic tile **134** when the thermal protection apparatus **130** is heated results in the top portions of the edgings **137** (i.e., the portions of the edgings **137** near the outer metal layer **133**) expanding and deforming significantly more than the bottom portions of the edgings **137** (i.e., the portions of the edgings **137** near the inner metal layer **135**), which may not substantially deform or expand due to heat.

(31) Each of the thermal protection apparatuses **130** can include the lip portion **139**. For example, in some embodiments, each of the generally rectangular thermal protection apparatuses **130** can have two adjacent sides that each includes the lip portion **139** while the other two sides do not. In these embodiments, the thermal protection apparatuses **130** are arranged such that the edges having the lip portion **139** are positioned directly adjacent to the edges of an adjacent thermal protection apparatus **130** that do not have the lip portion **139**. In this way, the thermal protection apparatuses **130** can be arranged such that each gap **148** is covered by a single lip portion **139**. In other embodiments, the thermal protection apparatuses **130** can include lip portions **139** on two opposing edges and not the other two edges. In still other embodiments, some of the thermal protection apparatuses **130** that form the heat shield can include lip portions **139** on all four edges while other thermal protection apparatuses **130** do not include lip portions **139** along any of the edges.

(32) In some embodiments, none of the thermal protection apparatuses **130** include lip portions **139**. In these embodiments, the gaps **148** can be sealed using other sealing mechanisms. FIG. 8 shows a cross-sectional view of a representative sealing mechanism. In this embodiment, neither of the outer metal layers **133** of adjacent thermal protection apparatuses have lip portions that extend over the gap **148**. Instead, a T-seal **151** is positioned within the gap **148**. The T-seal **151** can be a generally T-shaped structure having a first portion **152** coupled to a second portion **153** that is generally perpendicular to the first portion **152**. The first portion **152** can be positioned on the outer surface **131** and can extend across the gap **148** while the second portion **153** can be positioned within the gap **148** and can extend from the first portion **152** to the body **115** of the launch vehicle. The T-seal **151** can be formed from metal and can be welded to the thermal protection apparatuses **130** and/or the body **115** to seal the gap **148**. For example, the first portion **152** can be welded (e.g., tack welded) to the outer metal layers **133** of the two adjacent thermal protection apparatuses **130** and the second portion **153** can be welded to the body **115** within the gap **148**. In this way, the T-seal **151** can help to prevent or inhibit heat and water from entering the gap **148**. In some embodiments, ceramic gaskets can be positioned within the gap **148** between the second portion **153** and the edgings **137** in order to provide additional thermal protection to the body **115**.

(33) In the illustrated embodiments, the outer metal layer **133** is formed from sheet metal cut to a suitable size and shape and positioned on the ceramic tile **134**. In other embodiments, the outer metal layer **133** can be formed using other techniques. For example, in some embodiments, the outer metal layer **133** can be formed using a thermal spray technique. In these embodiments, holes can be drilled into the ceramic tile **134** and the pins can be inserted into the holes before the outer metal layer **133** is formed. After depositing the pins within the holes, metal feedstock can be melted (e.g., via electricity, plasma, or a flame) and the molten metal can be sprayed or otherwise disposed over the top surface of the ceramic tile **134**. The molten metal can weakly bond (or not bond at all) with the ceramic tile **134** but can strongly bond with the metal pins.

(34) From the foregoing, it will be appreciated that several embodiments of the disclosed technology have been described herein for purposes of illustration, but that various modifications can be made without deviating from the technology. For example, in some applications, the thermal protection apparatus can be filled with a material other than a ceramic, and/or can be or can include a soft and flexible material. The thermal protection apparatus can be coupled to any portion of a

launch vehicle, and/or vehicles that do not ascend into space, such as airplanes and/or helicopters. The thermal protection apparatus can be applied to stationary structures such as furnaces and power plants. More generally, in some embodiments, the thermal protection apparatus can be coupled to any suitable structure to provide insulation to that structure. In some embodiments, a single thermal protection apparatus includes multiple pins to secure the outer and inner metal portions to the tile, and in some embodiments, a single pin performs this function.

(35) Certain aspects of the technology described in the context of particular embodiments can be combined or eliminated in other embodiments. For example, the edging can be positioned around the top half of the ceramic tile but not the bottom half or can be eliminated entirely. Further, while advantages associated with some embodiments of the disclosed technology have been described herein, configurations with different characteristics can also exhibit such advantages, and not all configurations need necessarily exhibit such advantages to fall within the scope of the technology. Accordingly, the disclosure and associated terminology can encompass other arrangements not expressly shown or described herein.

(36) To the extent any materials incorporated herein by reference conflict with the present disclosure, the present disclosure controls. As used herein, the phrase “and/or” as in “A and/or B” refers to A alone, B alone and both A and B.

Claims

1. A thermal protection apparatus, comprising: a rigid ceramic tile having a top surface, a bottom surface, and a plurality of holes that extend between the top surface and the bottom surface; an outer metal portion positioned adjacent to and fully covering the top surface, wherein the outer metal portion defines an exterior surface of the thermal protection apparatus; a corrugated metallic edging positioned around a perimeter of the ceramic tile, wherein the corrugated metallic edging has a top bent portion coupling the corrugated metallic edging to the outer metal portion; and a plurality of elongate elements positioned within corresponding ones of the holes, wherein individual ones of the elongate elements are coupled to the outer metal portion.
2. The thermal protection apparatus of claim 1, further comprising an inner metal portion positioned adjacent to and fully covering the bottom surface.
3. The thermal protection apparatus of claim 2 wherein individual ones of the elongate elements are coupled to the inner metal portion.
4. The thermal protection apparatus of claim 2 wherein the corrugated metallic edging is further coupled to the inner metal portion, and wherein the corrugated metallic edging is configured to remain coupled to the outer metal portion and the inner metal portion as the outer metal portion expands.
5. The thermal protection apparatus of claim 1 wherein the corrugated metallic edging extends between the top surface and the bottom surface.
6. The thermal protection apparatus of claim 1 wherein— the ceramic tile includes a notch formed in the top surface that extends around the perimeter of the ceramic tile, and the top bent portion is positioned in the notch such that the top bent portion is coplanar with the top surface and is positioned between the ceramic tile and the outer metal portion.
7. The thermal protection apparatus of claim 1 wherein the outer metal portion includes a lip portion that projects laterally past the corrugated metallic edging.
8. The thermal protection apparatus of claim 1 wherein the top bent portion is coplanar with the top surface and positioned between the ceramic tile and the outer metal portion.
9. The thermal protection apparatus of claim 1, further comprising an inner metal portion positioned adjacent to and fully covering the bottom surface, and wherein the corrugated metallic edging includes a bottom bent portion coupling the corrugated metallic edging to the inner metal portion.

10. The thermal protection apparatus of claim 9 wherein the ceramic tile includes a notch formed in the bottom surface and that extends around the perimeter of the ceramic tile, and wherein the bottom bent portion is positioned within the notch such that the bottom bent portion is coplanar with the bottom surface.
 11. The thermal protection apparatus of claim 1 wherein the holes are larger than the corresponding ones of the elongate elements such that the elongate elements can bend and flex within the corresponding ones of the holes.
 12. A thermal protection apparatus, comprising: a tile having a first surface, a second surface, and a plurality of holes that extend between the first surface and the second surface, wherein the tile is formed of a rigid material configured to inhibit heat transfer therethrough and to retain a shape and a size thereof when the tile is exposed to high temperature; a metal layer positioned adjacent to and fully covering the first surface, wherein the metal layer defines an exterior surface of the thermal protection apparatus; a corrugated metallic edging positioned around a perimeter of the tile, wherein the corrugated metallic edging has a top bent portion coupling the corrugated metallic edging to the metal layer; and a plurality of elongate elements positioned within corresponding ones of the holes, wherein individual ones of the elongate elements are coupled to the metal layer.
 13. The thermal protection apparatus of claim 12 wherein the corrugated metallic edging extends between the first surface and the second surface.
 14. The thermal protection apparatus of claim 12 wherein— the tile includes a notch formed in the first surface that extends around the perimeter of the tile, the top bent portion is positioned in the notch such that the top bent portion is coplanar with the first surface and is positioned between the tile and the metal layer, and the metal layer includes a lip portion that projects laterally past the corrugated metallic edging.
 15. The thermal protection apparatus of claim 12 wherein the metal layer is a first metal layer, and further comprising a second metal layer positioned adjacent to and fully covering the second surface.
 16. A thermal protection apparatus, comprising: a rigid ceramic tile having a top surface and a bottom surface; an outer metal layer positioned adjacent to and fully covering the top surface, wherein the outer metal layer defines an exterior surface of the thermal protection apparatus; and an edging extending between the top surface and the bottom surface and positioned around a perimeter of the ceramic tile, wherein the edging has a top bent portion coupling the edging to the outer metal layer and the edging is formed of a corrugated metal.
 17. The thermal protection apparatus of claim 16, further comprising at least one pin extending through the ceramic tile and coupled to the outer metal layer.
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